DO NOW — RECALL

Lesson 111 · OS Types, Boot Firmware, Drivers and VMs

Name: _____ Class: _____ Date: _____

1. In round robin, what happens when a running process's quantum expires but it is unfinished?

2. After pushing A then B onto a stack, which is popped first?

3. Why must a binary search operate on ordered data?

4. An OR gate has inputs 0 and 1. What is the output?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It returns to the ready queue (subject to scheduler rules).

2. Model answer

B, because a stack is LIFO.

3. Model answer

It discards half the search space using an ordering comparison.

4. Model answer

1.

By the end, you can…

• Embedded/real-time emphasise constraints; multi-user and distributed coordinate shared work.

• BIOS/UEFI firmware begins startup; drivers provide hardware-specific access.

• A hypervisor runs guest OSs; a language VM executes intermediate code.

Vocabulary — know these terms

real-time OS — System where meeting time constraints is essential.

BIOS/UEFI — Platform firmware that begins boot preparation.

hypervisor — Manages virtual machines running guest OSs.

device driver — Hardware-specific interface supplied to an OS.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

Use a VM with a hypervisor to run Linux as a guest OS, while a printer driver exposes the printer to the host OS or guest when assigned. Boot firmware starts hardware initialisation before the main OS loads.

IMPROVE: Explain one correction to your own reasoning.

NEW LEARNING

Read, annotate and explain

Match OS to use

Embedded/real-time emphasise constraints; multi-user and distributed coordinate shared work.

Boot and control

BIOS/UEFI firmware begins startup; drivers provide hardware-specific access.

Virtualise

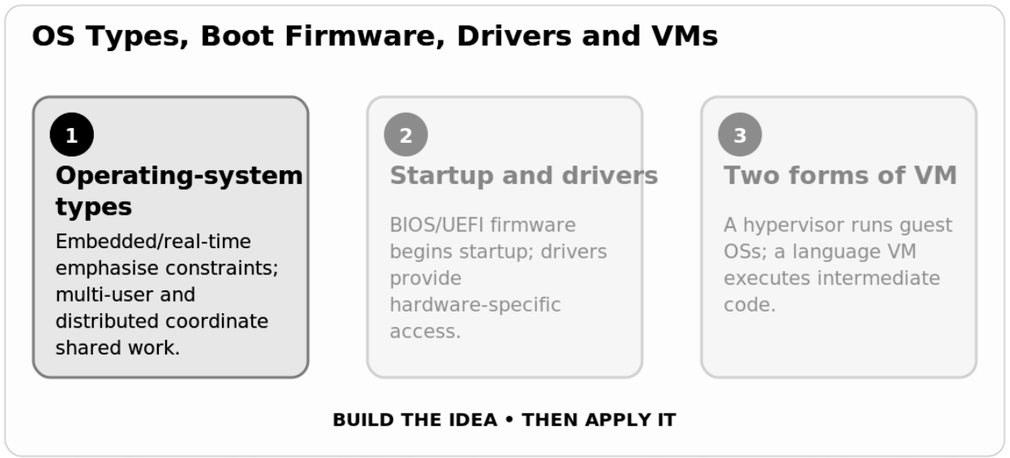
A hypervisor runs guest OSs; a language VM executes intermediate code.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

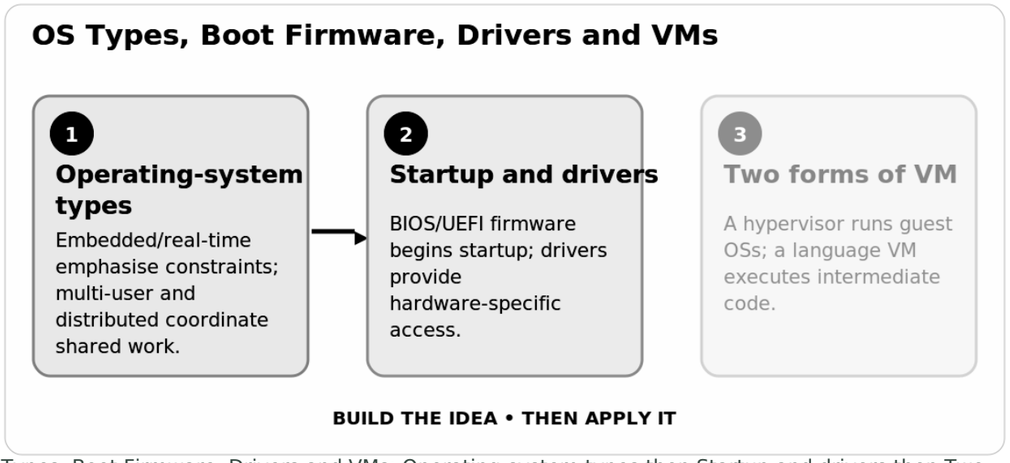
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



OS Types, Boot Firmware, Drivers and VMs: Operating-system types then Startup and drivers then Two forms of VM

New Learning



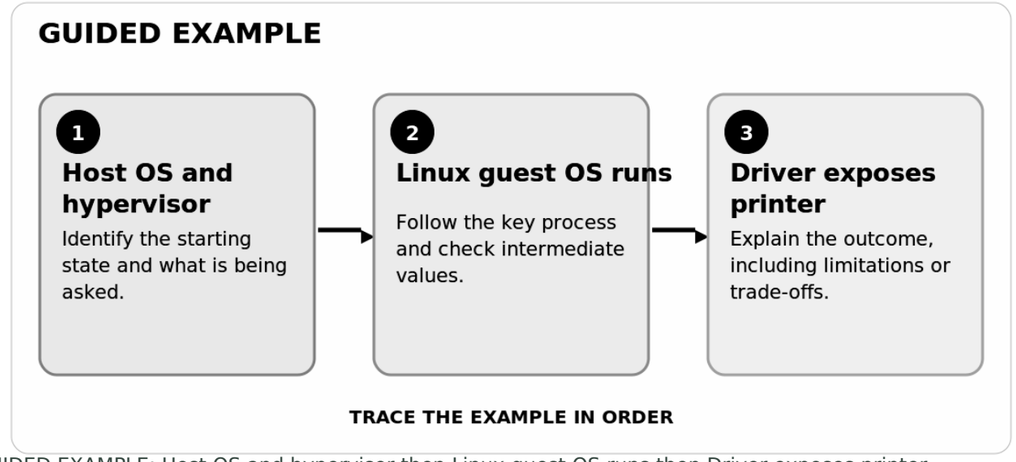
OS Types, Boot Firmware, Drivers and VMs: Operating-system types then Startup and drivers then Two forms of VM

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school wants to test Linux on Windows hardware and use a new printer. Explain the system software involved.

Identify the inputs, relevant facts and exact output or judgement required.



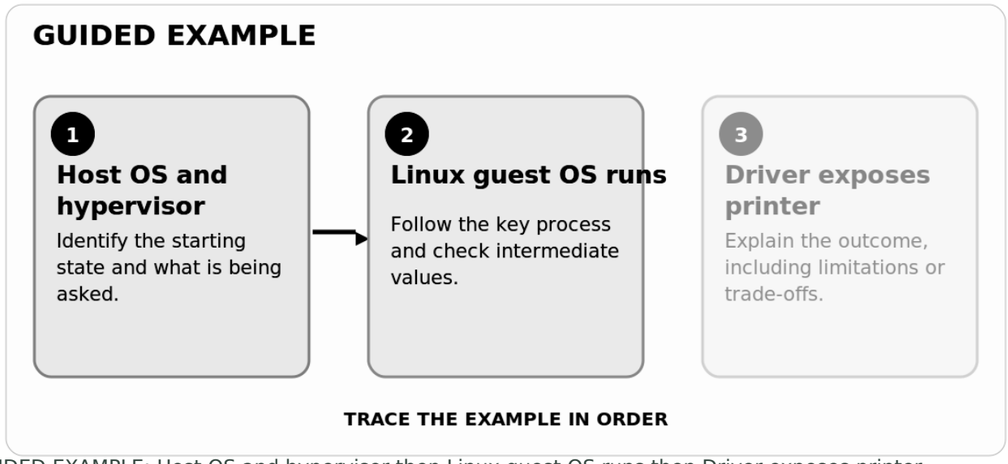
GUIDED EXAMPLE: Host OS and hypervisor then Linux guest OS runs then Driver exposes printer

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school wants to test Linux on Windows hardware and use a new printer. Explain the system software involved.

Identify the inputs, relevant facts and exact output or judgement required.

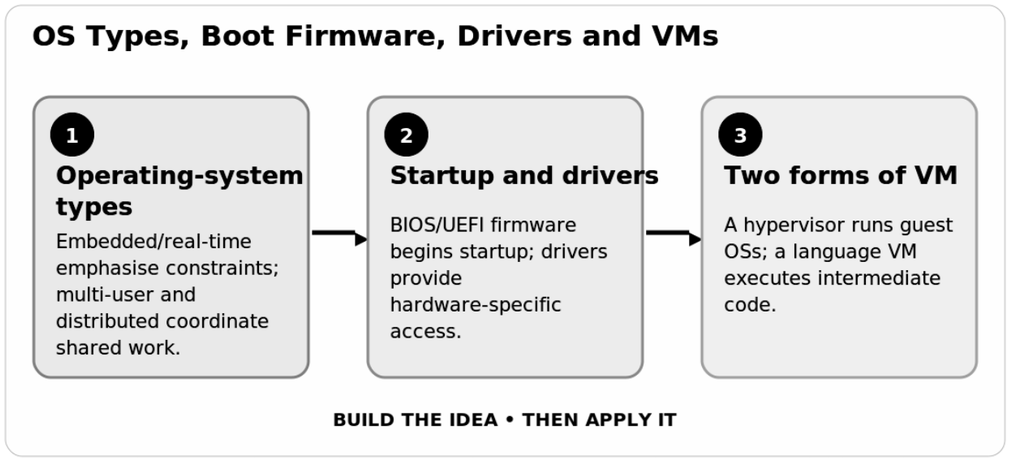


GUIDED EXAMPLE: Host OS and hypervisor then Linux guest OS runs then Driver exposes printer

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



OS Types, Boot Firmware, Drivers and VMs: Operating-system types then Startup and drivers then Two forms of VM

I DO — WORKED EXAMPLE

Follow the worked method

An industrial safety controller must respond within a guaranteed deadline. Choose an OS type and justify it.

Model answer

A hard real-time operating system is suited because missing a deadline is unacceptable; deterministic scheduling/latency matters more than high average throughput. An embedded implementation may also apply.

Check: link each claim to the scenario and use precise OCR terminology.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Every virtual machine must run a complete guest operating system.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Compare a hypervisor virtual machine with a bytecode virtual machine and explain a real-time deadline. [8]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

An industrial safety controller must respond within a guaranteed deadline. Choose an OS type and justify it.

A hard real-time operating system is suited because missing a deadline is unacceptable; deterministic scheduling/latency matters more than high average throughput. An embedded implementation may also



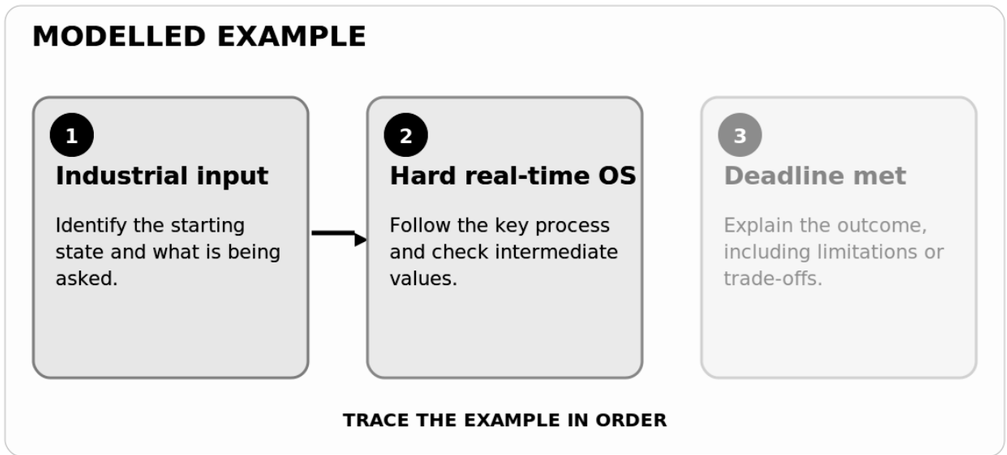
MODELLED EXAMPLE: Industrial input then Hard real-time OS then Deadline met

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

An industrial safety controller must respond within a guaranteed deadline. Choose an OS type and justify it.

A hard real-time operating system is suited because missing a deadline is unacceptable; deterministic scheduling/latency matters more than high average throughput. An embedded implementation may also



MODELLED EXAMPLE: Industrial input then Hard real-time OS then Deadline met

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Explain two differences between multi-user and multitasking operating systems, giving examples. [6]

Question 2

Describe the roles of BIOS/UEFI firmware and a device driver during startup and use. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: What must we decide? Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

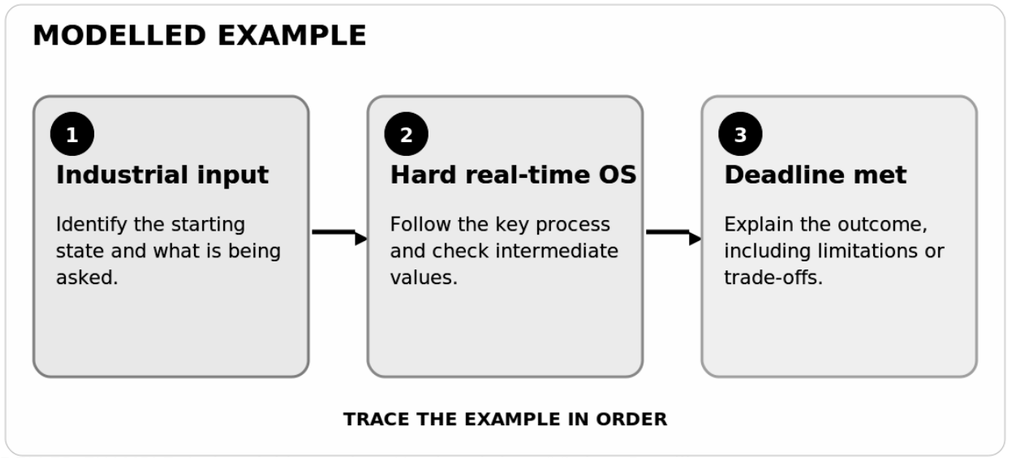
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

An industrial safety controller must respond within a guaranteed deadline. Choose an OS type and justify it.

A hard real-time operating system is suited because missing a deadline is unacceptable; deterministic scheduling/latency matters more than high average throughput. An embedded implementation may also



MODELLED EXAMPLE: Industrial input then Hard real-time OS then Deadline met

WE DO — GUIDED PRACTICE

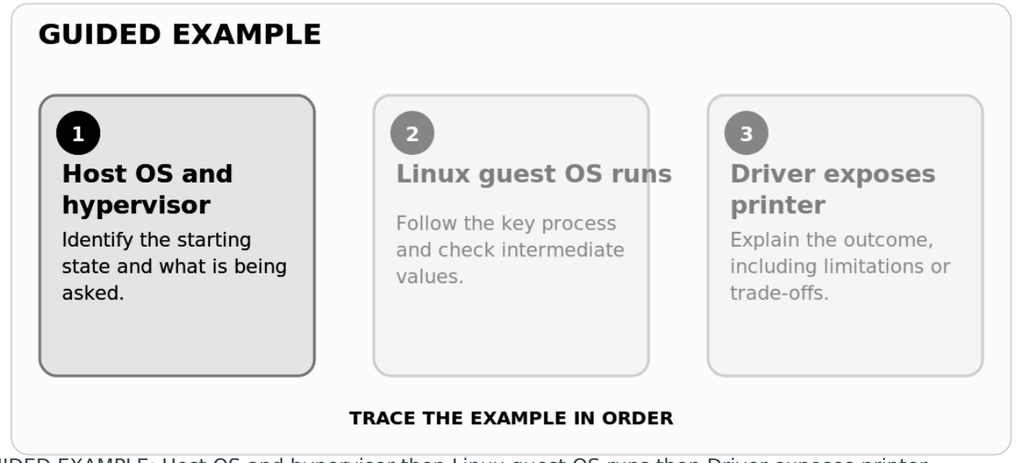
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school wants to test Linux on Windows hardware and use a new printer. Explain the system software involved.

Identify the inputs, relevant facts and exact output or judgement required.



GUIDED EXAMPLE: Host OS and hypervisor then Linux guest OS runs then Driver exposes printer